THE REAL-ROOTEDNESS PROBABILITY OF A RANDOM CUBIC POLYNOMIAL: FIVE CLOSED FORMS

VÍT PLÁNOČKA

Abstract. We compute, in closed form, the probability that a random cubic polynomial has three real roots, in five related models. For a *monic* cubic $x^3 + ax^2 + bx + c$ with (a, b, c) independent and uniform on $[-1, 1]$, this probability is $\frac{383}{4860} + \frac{\ln 3}{48}$; on $[0, 1]$ it is exactly $\frac{1}{2880}$. For the *non-monic* cubic $ax^3 + bx^2 + cx + d$ with all four coefficients independent and uniform on $[-1, 1]$, it is $\frac{641}{2430} - \frac{\ln 3}{24}$, and on $[0, 1]$ it is $\frac{719}{2880} - \frac{\ln 2}{3}$ – the four uniform values completing a 2×2 table over $\{\text{monic}, \text{non-monic}\} \times \{[-1, 1], [0, 1]\}$. For the monic cubic with (a, b, c) i.i.d. *standard normal*, it is a single one-dimensional integral of elementary functions, Theorem 4, obtained via a Kac–Rice level-crossing argument with an exact Owen's-T-function cancellation. Four of the five values appear to be new; the remaining one (non-monic, $[-1, 1]$) confirms, with a first proof, a closed form that had circulated unproved in an informal online forum thread since June 2026. All five results are verified by independent symbolic and numerical computation and by a complete, zero-`sorry` machine-checked proof in Lean 4 / Mathlib – which along the way proves, from scratch, the classical fact connecting a real cubic's discriminant sign to its real-root count (seemingly absent from Mathlib until now), and formalizes the Gaussian case's entire probabilistic argument without appeal to any general Rice's formula, coarea formula, or theory of stochastic processes – none of which Mathlib has – by exploiting that the relevant process is really just a fixed three-dimensional Gaussian vector composed with a deterministic curve.

Contents

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With assistance from Claude (Anthropic). Machinery, proofs, and their verification (symbolic, numerical, and machine-checked in Lean 4 / Mathlib) are recorded in the accompanying repository.

1. Introduction

How likely is a "random" polynomial to have all of its roots real? For degree 2 this is classical. For a monic quadratic $x^2 + bx + c$ with (b, c) uniform on $[-1, 1]^2$, real roots occur exactly when $c \le b^2/4$, giving probability $13/24$; for the full quadratic ax^2+bx+c the answer involves a logarithm, $\frac{41}{72} + \frac{\ln 2}{12}$ on $[-1, 1]^3$ and $\frac{5}{36} + \frac{\ln 2}{6}$ on $[0, 1]^3$, the latter apparently first recorded (without derivation) by J. D'Aurizio and rederived in [3]. For the analogous *cubic* question the literature is far thinner. The only paper we could locate that claims exact results for random cubics with general coefficient distributions is Li [1], which is effectively unobtainable (two citations in thirty-eight years, neither reproducing a value) and, per the recovered *Mathematical Reviews* notice [2], treats a family of symmetric-interval distributions that does not include the one-sided case of our Theorem 2 at all, and reaches our Theorem 1 only as an unverifiable special case.

This note settles five instances of the cubic question completely.

Theorem 1 (Monic cubic, symmetric cube). *Let (a, b, c) be independent and uniform on $[-1, 1]$. Then*

$$\mathbb{P}\big(x^3 + ax^2 + bx + c \textit{ has three real roots}\big) = \frac{383}{4860} + \frac{\ln 3}{48} = 0.1016943403760588695995408 6\ldots$$

Theorem 2 (Monic cubic, unit cube). *Let (a, b, c) be independent and uniform on $[0, 1]$. Then*

$$\mathbb{P}\big(x^3 + ax^2 + bx + c \textit{ has three real roots}\big) = \frac{1}{2880}.$$

Theorem 3 (Non-monic cubic). *Let (a, b, c, d) be independent and uniform on $[-1, 1]$. Then*

$$\mathbb{P}\big(ax^3+bx^2+cx+d \textit{ has three real roots}\big) = \frac{641}{2430} - \frac{\ln 3}{24} = 0.2180104962026147710889841233586 8\ldots$$

Theorem 4 (Gaussian cubic). *Let (a, b, c) be independent, standard normal. Then*

$$\mathbb{P}\big(x^3 + ax^2 + bx + c \textit{ has three real roots}\big) = \frac{1}{\pi}\int_0^\infty e^{-\frac{x^4(x^4+4x^2+9)}{2(x^4+4x^2+1)}} \frac{2(x^4 + 6x^2 + 3)}{\sqrt{x^4 + 4x^2 + 1}\,(x^4 + 4x^2 + 9)}\, dx$$
$$= 0.1699293826234795026564431571 32\ldots$$

Theorem 5 (Non-monic cubic, unit cube). *Let (a, b, c, d) be independent and uniform on $[0, 1]$. Then*

$$\mathbb{P}\big(ax^3+bx^2+cx+d \textit{ has three real roots}\big) = \frac{719}{2880} - \frac{\ln 2}{3} = 0.0186037175911293413053670706251 0\ldots$$

The four uniform results, Theorems 1–3 and 5, exhaust the natural 2×2 grid of {monic, non-monic} against {symmetric, one-sided} coefficient cubes:

	uniform $[-1, 1]$	uniform $[0, 1]$
monic $x^3 + ax^2 + bx + c$	$\frac{383}{4860} + \frac{\ln 3}{48}$	$\frac{1}{2880}$
non-monic $ax^3 + bx^2 + cx + d$	$\frac{641}{2430} - \frac{\ln 3}{24}$	$\frac{719}{2880} - \frac{\ln 2}{3}$

The two one-sided values carry $\ln 2$ and the two symmetric values $\ln 3$; in Theorem 5 a $\ln 3$ does appear mid-derivation but cancels between the two branches of the reduced integral, leaving only $\ln 2$ – the same constant as the one-sided *quadratic* $\frac{5}{36} + \frac{\ln 2}{6}$.

Theorems 1, 2, 4 and 5 are, to our knowledge, new: an eight-agent literature sweep together with a targeted search for the exact constants (Section 7) found no prior publication of any of the four values, digit strings, or exact expressions. Theorem 3 is more subtle: the exact value $\frac{641}{2430} - \frac{\ln 3}{24}$ already existed, unproved, as the terminus of a long, informally written, partly AI-assisted derivation posted to a Russian mathematics forum (`dxdy.ru`) in June 2026; Section 4 gives the first proof.

Theorem 4 is qualitatively different from the four uniform cases: with no cube to bound anything, the elementary band/discriminant argument behind them does not apply in any form, and the natural object is instead a probabilistic level-crossing count. Its "closed form" is a single definite integral of elementary functions – the same sense in which the error function or Owen's T-function are accepted as closed forms in the literature – rather than a finite combination of named constants; Section 5 gives the proof.

Motivation. The cubic question is the tractable sibling of a harder open problem: for an $n \times n$ matrix with i.i.d. entries, what is the probability that all n eigenvalues are real? This is classical for $n = 2$ but open for $n \geq 3$, both for uniform entries (no known closed form, no nontrivial rigorous bound; the canonical Mathematics Stack Exchange statement of the question has 153 upvotes and zero answers [6]) and, in the Gaussian case, for anything beyond Edelman's exact eigenvalue-count formula $2^{-n(n-1)/4}$ [5]. Moving from matrix-entry space to *coefficient* space – the probability that a random *characteristic polynomial* is real-rooted – keeps the same underlying object (the measure of a region cut out by a discriminant) while dropping the ambient dimension from 9 to 3–4, where it becomes tractable. This note attacks that lower-dimensional sibling in both the uniform and Gaussian coefficient models.

Machine verification. Every calculus step below is checked twice over in exact arithmetic (once by hand, once independently by computer algebra), confirmed by high-precision quadrature and direct Monte Carlo sampling, and – new for a paper at this level of the literature – verified by a *complete, zero-`sorry` machine-checked proof of all four theorems* in Lean 4 against Mathlib, described in Section 6. That formalization also produced two results of independent interest: a from-scratch, calculus-free proof that a real cubic's discriminant sign determines its real-root count (a classical fact we could not find already in Mathlib), and a complete formalization of Theorem 4's probabilistic argument that needs none of Rice's formula, the coarea formula, or a general theory of stochastic processes – Mathlib has none of them – built instead from first principles for this specific finite-dimensional Gaussian family.

2. Preliminaries

Throughout, Δ denotes the discriminant of a cubic. For the general cubic $ax^3 + bx^2 + cx + d$,

$$\Delta(a, b, c, d) = 18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2, \tag{1}$$

and classically, for $a \neq 0$: $\Delta > 0$ iff the cubic has three distinct real roots; $\Delta = 0$ iff it has a repeated root (all real); $\Delta < 0$ iff it has one real root and a complex-conjugate pair. For the monic case $a = 1$ we write $\Delta_3(b, c, d)$ for $\Delta(1, b, c, d)$; abusing notation slightly to match the variable names used in Theorems 1–2 (whose monic cubic is $x^3 + ax^2 + bx + c$, i.e. our $(b, c, d) \mapsto (a, b, c)$),

$$\Delta_3(a, b, c) = 18abc - 4a^3c + a^2b^2 - 4b^3 - 27c^2. \tag{2}$$

3. The monic cubic

3.1. **The band reduction.** Write $f(x) = x^3 + ax^2 + bx + c$ and $g(x) = x^3 + ax^2 + bx = f(x) - c$. For $b < a^2/3$, put $s = \sqrt{a^2 - 3b}$; the critical points of f (equivalently of g) are

$$x_- = \frac{-a-s}{3} \text{ (local max)}, \qquad x_+ = \frac{-a+s}{3} \text{ (local min)}.$$

f has three real roots iff $f(x_-) \geq 0 \geq f(x_+)$, i.e. iff c lies in the band

$$c \in [c_{\mathrm{lo}}, c_{\mathrm{hi}}], \qquad c_{\mathrm{lo}} = -g(x_-), \quad c_{\mathrm{hi}} = -g(x_+), \qquad c_{\mathrm{hi}} - c_{\mathrm{lo}} = \frac{4}{27}s^3.$$

This is not merely an implication but captures the discriminant exactly: as a polynomial in c,

$$\Delta_3(a,b,c) = -27\,(c - c_{\mathrm{lo}})(c - c_{\mathrm{hi}}). \tag{3}$$

(3) is verified by direct expansion (and, independently, by `sympy`). For $b > a^2/3$ the right-hand quadratic in c has no real roots and $\Delta_3 < 0$ identically (one real root). Hence for any c-window $[w_{\mathrm{lo}}, w_{\mathrm{hi}}]$,

$$\mathrm{vol} = \iint_{b<a^2/3} \mathrm{len}\big([c_{\mathrm{lo}}, c_{\mathrm{hi}}] \cap [w_{\mathrm{lo}}, w_{\mathrm{hi}}]\big)\, db\, da. \tag{4}$$

3.2. **Proof of Theorem 1.**

Lemma 6 (Never-clipped). *On $D = \{(a,b) : a \in [-1,1],\ b \in [-1, a^2/3]\}$, $c_{\mathrm{hi}} \leq 1$, with equality only at $(a,b) = (-1,-1)$.*

Proof. Since $\partial g/\partial a$ at fixed x equals x^2 and $g'(x_+) = 0$ kills the chain-rule term at the critical point, $\partial c_{\mathrm{hi}}/\partial a = -x_+^2 \leq 0$; so c_{hi} is non-increasing in a. As $b \leq a^2/3 \leq 1/3$ for $a \in [-1,1]$, the point $(-1,b)$ is also in D for every $(a,b) \in D$, so $c_{\mathrm{hi}}(a,b) \leq c_{\mathrm{hi}}(-1,b)$. On the edge $a = -1$: $\partial c_{\mathrm{hi}}/\partial b = -x_+$ with $x_+ = (1 + \sqrt{1-3b})/3 > 0$, so c_{hi} is strictly decreasing in b there, with maximum at $b = -1$: $c_{\mathrm{hi}}(-1,-1) = 1$ exactly. □

The map $(a,b,c) \mapsto (-a,b,-c)$ preserves Δ_3 (direct expansion) and the cube, and sends c_{hi} to $-c_{\mathrm{lo}}$; hence $c_{\mathrm{lo}} \geq -1$ on D as well. **The band never exits the window** $[-1,1]$: it touches the boundary only at the two corner points $(\mp 1, -1, \pm 1)$, of measure zero. Therefore, using (4) with no clipping correction,

$$\mathrm{vol} = \int_{-1}^{1}\int_{-1}^{a^2/3} \frac{4}{27}(a^2 - 3b)^{3/2}\, db\, da = \frac{8}{405}\int_{-1}^{1}(a^2+3)^{5/2}\, da = \frac{3064}{1215} + \frac{8}{3}\operatorname{arsinh}\frac{1}{\sqrt{3}}.$$

Since $\operatorname{arsinh}(1/\sqrt{3}) = \ln(1/\sqrt{3} + 2/\sqrt{3}) = \frac{1}{2}\ln 3$,

$$\mathbb{P} = \mathrm{vol}/8 = \frac{383}{4860} + \frac{\ln 3}{48}.$$

Remark 7. The corner-touching geometry is exactly why this is solvable in closed form: the naive "main term plus clipping corrections" program has *all* corrections identically zero, so the trivial upper bound already equals the answer. High-precision quadrature landing on that upper bound to sixteen digits was, in practice, the signal that this was happening.

3.3. **Proof of Theorem 2.** Now the domain is $a \in [0,1]$, $b \in [0, a^2/3]$, window $[0,1]$. Since $b \geq 0$ forces $s \leq a$, both critical points satisfy $x_- \leq 0 \leq$ (the argument of the local max) and $x_+ \leq 0$; and $c_{\mathrm{hi}} \geq 0$ because $g(0) = 0$, $g'(0) = b \geq 0$, and $x_+ \leq 0$ is the local minimum, so $g(x_+) \leq 0$. A short computation shows $c_{\mathrm{hi}} \leq 1/27$ throughout the domain, so the top of the window $[0,1]$ never binds. The bottom does: $\partial c_{\mathrm{lo}}/\partial b = -x_- \geq 0$, and $c_{\mathrm{lo}} = 0$ exactly on the curve $b = a^2/4$ (where $x_- = -a/2$ and $g(-a/2) = 0$); so $c_{\mathrm{lo}} \leq 0$ for $b \leq a^2/4$ and $c_{\mathrm{lo}} \geq 0$ for $a^2/4 \leq b \leq a^2/3$ – a single sign change. Hence the overlap length is c_{hi} on $[0, a^2/4]$ and the full band width on $[a^2/4, a^2/3]$, and, remarkably, all fractional powers cancel:

$$\int_0^{a^2/4} c_{\text{hi}}\, db + \int_{a^2/4}^{a^2/3} \frac{4}{27}(a^2-3b)^{3/2}\, db = \frac{a^5}{480}.$$

Integrating over $a \in [0,1]$,

$$\mathbb{P} = \int_0^1 \frac{a^5}{480}\, da = \frac{1}{2880}.$$

4. The non-monic cubic

Dropping the monic restriction – letting the leading coefficient a range uniformly over $[-1,1]$, including near 0 – makes the problem genuinely four-dimensional, and Lemma 6's trick does not generalize directly: it is special to a symmetric window of half-width *exactly* 1 around a *fixed* leading coefficient.

Writing $V(t)$ for the volume of the real-rooted region inside the cube $[-t,t]^3$ in the monic-cubic coefficient space (so $V(1) = 8 \cdot \mathbb{P}$ of Theorem 1), the natural attack conditions on the leading coefficient and rescales:

$$\mathbb{P}(\text{non-monic}) = \int_0^1 \frac{a^3}{8}\, V(1/a)\, da. \tag{5}$$

This requires $V(t)$ for every $t \geq 1$, and (5) turns out to be a dead end: high-precision numerics for $p(a) := a^3 V(1/a)/8$ (computed to 40 digits; e.g. $p(1/2) = 0.1732556347784\ldots$ and $p(1/3) = 0.2277312312100\ldots$) show no recognizable closed form under a blind PSLQ search against $\{1, \ln 2, \ln 3, \ln 5, \sqrt{2}, \sqrt{3}, \pi\}$. $V(1)$ is elementary only because $t = 1$ is exactly where the corner-touching geometry of Lemma 6 applies – not because $V(t)$ is nice in general.

4.1. **The discriminant region is a cone.**

The route that works is structural rather than a direct generalization of the band calculus. The discriminant (1) is homogeneous of degree 4: $\Delta(\lambda x) = \lambda^4 \Delta(x)$ for $x = (a,b,c,d)$, so the real-rooted set $R = \{\Delta > 0\}$ is invariant under $x \mapsto \lambda x$ $(\lambda > 0)$ – a cone. Apply the divergence theorem to $F = x/4$ (so $\operatorname{div} F = 1$) on $R \cap [-1,1]^4$. Euler's identity gives $x \cdot \nabla\Delta = 4\Delta$, which vanishes identically on the lateral boundary $\{\Delta = 0\}$; so that part of the surface integral is zero, and only the eight faces of the cube contribute:

$$\operatorname{vol}_4(R \cap [-1,1]^4) = \tfrac{1}{4} \oint_{\partial(R\cap[-1,1]^4)} F \cdot n\, dS = \tfrac{1}{2}(S_a + S_b + S_c + S_d),$$

where S_a, S_b, S_c, S_d are the (signed, outward) contributions of the four faces $a = \pm 1, \ldots$ paired by central symmetry $x \mapsto -x$ (which preserves Δ and the cube). The coefficient reversal $(a,b,c,d) \mapsto (d,c,b,a)$ – sending each root x to its reciprocal $1/x$ – also preserves both R and the cube, and identifies $S_a = S_d$, $S_b = S_c$. Hence

$$\operatorname{vol}_4(R \cap [-1,1]^4) = V(1) + S_b, \qquad \mathbb{P}(\text{non-monic}) = \frac{V(1) + S_b}{16}, \tag{6}$$

where $V(1) = S_a = \frac{766}{1215} + \frac{\ln 3}{6}$ (eight times Theorem 1) and

$$S_b = \operatorname{vol}_3\bigl\{(a,c,d) \in [-1,1]^3 : ax^3 + x^2 + cx + d \text{ has three real roots}\bigr\}$$

is the volume on the face where the *second* coefficient is pinned to 1 – a genuinely new, but purely three-dimensional, problem, with neither a $t \to \infty$ tail nor an $a \to 0$ endpoint singularity to contend with.

Remark 8. We verified the reduction step (6) itself against a case with an independently known answer: applied to $q^2 > 4pr$ on $[-1,1]^3$ (also a cone, the quadratic discriminant being homogeneous), the three faces give $S_p = S_r = 13/6$, $S_q = 5/2 + \ln 2$, and $\frac{1}{3} \cdot 2(S_p + S_q + S_r)$ works out to $\frac{41}{9} + \frac{2}{3}\ln 2$ – exactly eight times the known full-quadratic volume $\frac{41}{72} + \frac{\ln 2}{12}$. The reduction is an identity, not an approximation.

4.2. **Computing S_b.** Substitute $s = \sqrt{1 - 3ac}$ (so $c = (1 - s^2)/(3a)$). The admissible-d band on this face is $[-K_-/u,\ K_+/u]$ with

$$K_+ = (s-1)^2(2s+1), \qquad K_- = (s+1)^2(2s-1), \qquad u = 27a^2, \qquad K_+ + K_- = 4s^3,$$

on the domain $\{s \in (0,2),\ a \in [|s^2-1|/3, 1]\}$. Integrating over a *first*, at fixed s, is elementary, since a enters only through $u = 27a^2$; two lemmas make it precise.

Lemma 9 (Never clips above)**.** $d_{\mathrm{hi}} \le 1$ *throughout:* $\alpha_+ := \sqrt{K_+/27} < a_0 := |s^2-1|/3$ *reduces to* $2s+1 < 3(s+1)^2$*, true for all* s*. This is the exact analogue of Lemma 6.*

Lemma 10 (Clips below exactly for $s \in (2/3, 2)$)**.** $\alpha_- := \sqrt{K_-/27} > a_0$ *reduces to* $(3s-2)(s-2) < 0$*; and* $\alpha_- < 1$ *reduces to* $(s+1)^2(2s-1) < 27$*, i.e.* $s < 2$*, with equality precisely at* $s = 2$*.*

With Lemmas 9–10, $F(s) := \int_{a_0}^{1} L(a,s)/a\, da$ (the a-integral of the band length) is elementary – the term $-K_+/(2K_-) + 2s^3/K_-$ collapses to the constant $1/2$ via $K_+ + K_- = 4s^3$ – giving the piecewise closed form

$$F(s) = \begin{cases} \dfrac{2s^3}{27}\left(\dfrac{1}{a_0^2} - 1\right), & s \le 2/3, \\[2ex] \dfrac{K_+}{54a_0^2} + \dfrac{1}{2} + \ln\dfrac{\alpha_-}{a_0} - \dfrac{2s^3}{27}, & 2/3 < s < 2, \end{cases}$$

continuous at $s = 2/3$ (both branches equal 11264/18225 there). Then

$$S_b = \frac{4}{3}\int_0^2 s\,F(s)\,ds = \frac{1454}{405} - \frac{5}{6}\ln 3 = 2.674613216233365\ldots,$$

and, substituting into (6) with $V(1) = \frac{766}{1215} + \frac{\ln 3}{6}$,

$$\mathbb{P}(\text{non-monic}) = \frac{1}{16}\left(\frac{766}{1215} + \frac{\ln 3}{6} + \frac{1454}{405} - \frac{5}{6}\ln 3\right) = \frac{641}{2430} - \frac{\ln 3}{24}.$$

(Confirmed to hold exactly by direct symbolic simplification in `sympy`.)

4.3. **Independent verification.** Beyond exact symbolic confirmation (`sympy` verifies $\mathbb{P} - (\frac{641}{2430} - \frac{\ln 3}{24}) = 0$ identically), two checks are fully independent of the derivation above. A numerical route sharing no step with it – fixing the leading coefficient and integrating it out *last* rather than first, i.e. a direct high-precision evaluation of (5) – reproduces the exact value to *nineteen* significant digits; and blind PSLQ, given only the 22-digit numeric value of S_b and no closed form, recovers $S_b = \frac{1454}{405} - \frac{5}{6}\ln 3$ unprompted. Monte Carlo on the raw sign of (1), using no theory from this paper, gives $0.218008985 \pm 2.1 \times 10^{-6}$ at 4×10^{10} samples (-0.73σ).

That second route is delicate: the conditional probability given the leading coefficient has a square-root endpoint singularity as $a \to 0$, and the inner integral's clipping breakpoints both migrate to scale $\sqrt{3a}$ and are lost to floating point there. All three effects are invisible to a quadrature routine's own convergence estimate, and all had to be handled explicitly to reach nineteen digits.

The `dxdy.ru` thread's stated intermediate volume $V_0 = \frac{766}{1215} + \frac{\ln 3}{6}$ agrees with $V(1) = 8\cdot$ Theorem 1, and its final claimed value is confirmed exactly by Theorem 3; nothing here validates the specific route the thread took, and the derivation above is independent of it.

4.4. **The one-sided cube: proof of Theorem 5.** The same cone reduction applies over $[0,1]^4$, and reuses the face computation of §4 almost verbatim – with two changes, each of which makes the calculation *simpler* than its $[-1,1]$ counterpart.

The discriminant (1) is still homogeneous of degree 4, so $R = \{\Delta > 0\}$ is a cone; slicing $[0,1]^4$ by which coordinate is largest and using $\mathrm{vol}_3(t \cdot E) = t^3 \,\mathrm{vol}_3(E)$, $\int_0^1 t^3\,dt = \frac{1}{4}$, gives $\mathrm{vol}_4(R \cap [0,1]^4) = \frac{1}{4}(F_a + F_b + F_c + F_d)$, where F_j is the three-dimensional volume of the real-rooted region on the face $x_j = 1$. **The first change: four faces, not eight**, because on the positive orthant $\max_j |x_j| = \max_j x_j$, so the central symmetry $x \mapsto -x$ of §4 contributes nothing. The face $a = 1$ is the monic cubic $x^3 + bx^2 + cx + d$ on $[0,1]^3$ – exactly Theorem 2 – so $F_a = \frac{1}{2880}$; coefficient reversal $(a,b,c,d) \mapsto (d,c,b,a)$ gives $F_d = F_a$ and $F_c = F_b$. Hence

$$\mathbb{P}(\text{non-monic}, [0,1]) = \frac{1}{4}\Big(2 \cdot \tfrac{1}{2880} + 2F_b\Big) = \frac{1}{5760} + \frac{S}{2}, \qquad S := F_b. \tag{7}$$

S is the volume on the face $b = 1$, i.e. the same face integral as S_b of §4 – the polynomial $ax^3 + x^2 + cx + d$ with band $[-K_-/u, K_+/u]$, $u = 27a^2$, $K_\pm$ and $s = \sqrt{1-3ac}$ exactly as there – but now with $(a,c,d) \in [0,1]^3$. **The second change: s ranges only over** $(0,1]$, since $a, c \geq 0$ forces $ac \geq 0$; and the admissible-d window is $[0,1]$ rather than $[-1,1]$. Lemma 9 (top never clips) holds unchanged. The bottom of the band now clips against $d = 0$, not $d = -1$: since $d_{\mathrm{lo}} = -K_-/u$ and $K_- = (s+1)^2(2s-1)$, one has $d_{\mathrm{lo}} < 0$ exactly for $s > \frac{1}{2}$. So the clipped band length is $L = C(s)/(27a^2)$ with

$$C(s) = K_+ + \min(K_-, 0) = \begin{cases} 4s^3, & s \leq \frac{1}{2}, \\ K_+ = (s-1)^2(2s+1), & \frac{1}{2} < s \leq 1. \end{cases}$$

Because the floor is exactly 0 – where c_{lo} itself vanishes, an algebraic locus – the $(s-1)^2$ factor of K_+ cancels the $(1-s^2)^{-2}$ pole of the domain, and the inner a-integral $F(s) = \int_{a_0(s)}^1 L/a\,da = \frac{C(s)}{54}\big(9/(1-s^2)^2 - 1\big)$, $a_0(s) = (1-s^2)/3$, is a *rational* function of s with no logarithm at all (contrast (6)'s $F(s)$, which carries an $\ln(\alpha_-/a_0)$). With the Jacobian $dc = -(2s/3a)\,ds$,

$$S = \frac{2}{3}\int_0^1 s\,F(s)\,ds = \frac{2}{3}\Big(\underbrace{\tfrac{1199}{2160} - \tfrac{\ln 3}{2}}_{\int_0^{1/2}} + \underbrace{\tfrac{3341}{17280} + \tfrac{\ln 3}{2} - \ln 2}_{\int_{1/2}^1}\Big) = \frac{479}{960} - \frac{2}{3}\ln 2.$$

The $\ln 3$ produced by the $s \leq \frac{1}{2}$ branch is cancelled exactly by the $s > \frac{1}{2}$ branch, leaving only $\ln 2$. Substituting into (7),

$$\mathbb{P}(\text{non-monic}, [0,1]) = \frac{1}{5760} + \frac{1}{2}\Big(\frac{479}{960} - \frac{2}{3}\ln 2\Big) = \frac{719}{2880} - \frac{\ln 2}{3}.$$

This closed form was checked by nine mutually independent routes before being formalized: a raw 2×10^9-sample discriminant-sign Monte Carlo (0.05σ); an independent root-counting Monte Carlo that never forms Δ (cross-checked per-sample against companion-matrix eigenvalues, and re-run on the three known cells, all reproduced); a reduction-free randomized QMC of the full four-dimensional volume (-0.50σ); a second symbolic derivation with the opposite integration order (identical closed form); a 40-digit substitution-free quadrature of S agreeing to 2.5×10^{-39}; a deterministic quadrature confirming $F_a = \frac{1}{2880}$ with residual exactly 0 (decisively fixing the cone/face reduction); and blind `PSLQ` on both S and P, each returning the exact relation with a *zero* coefficient on $\ln 3$ – an independent confirmation that the $\ln 3$ genuinely cancels rather than being assumed away.

5. The Gaussian cubic

With (a, b, c) i.i.d. standard normal there is no cube, so nothing bounds the band and Lemma 6 has no analogue. The route that works counts *level crossings* instead of measuring a region.

5.1. **Reduction to a crossing count.** Write $f(x) = x^3 + ax^2 + bx + c$. Since f has odd degree it always has at least one real root, and it has three (counted without multiplicity, off the null set $\{\Delta_3 = 0\}$) exactly when it has a downward crossing — a point where f vanishes with $f' < 0$. Writing N for the number of real roots, $N \in \{1, 3\}$ almost surely, so with p the probability of three real roots,

$$\mathbb{E}[N] = 1 \cdot (1 - p) + 3 \cdot p = 1 + 2p, \qquad \text{i.e.} \qquad p = \tfrac{1}{2}\big(\mathbb{E}[N] - 1\big). \tag{8}$$

The virtue of (8) is that $\mathbb{E}[N]$, unlike p, is an integral of a *local* quantity and can be computed by a Kac–Rice argument.

5.2. **The Kac–Rice step.** Set $q(x) = x^4 + x^2 + 1$ and $u(x) = (x^2, x, 1)/\sqrt{q(x)}$, a unit vector, so that $Z(x) := (a, b, c) \cdot u(x)$ is standard normal for each fixed x and $f(x) = \sqrt{q(x)}\, Z(x) + x^3$. The pair $(Z(x), Z'(x))$ is a linear image of the single fixed Gaussian vector (a, b, c) – this is what makes the argument elementary, and is the reason no theory of stochastic processes is needed anywhere. A direct computation gives $\operatorname{Cov}(Z, Z') = 0$ (so the two are independent at each fixed x), together with

$$v(x)^2 := \operatorname{Var}(Z'(x)) = \frac{x^4 + 4x^2 + 1}{q(x)^2}, \qquad h(x) := \frac{x^3}{\sqrt{q(x)}}, \qquad h'(x) = \frac{x^2(x^4 + 2x^2 + 3)}{q(x)^{3/2}}.$$

Counting crossings of f and integrating the resulting local density over x yields, after evaluating the inner two-dimensional Gaussian integral in closed form,

$$\mathbb{E}[N] = \int_{\mathbb{R}} \varphi(h)\Big[2v\,\varphi(z) + h'\big(2\Phi(z) - 1\big)\Big]dx, \qquad z := h'/v, \tag{9}$$

with φ, Φ the standard normal density and distribution function.

5.3. **The Owen's-T cancellation.** Formula (9) still contains a normal CDF, and integrating it directly is unpromising. The key is that the Φ term is an exact derivative. With $\theta := \arctan(z/h)$ and T Owen's T-function, put

$$\Psi(x) := \tfrac{1}{2}\big(2\Phi(h) - 1\big) + 2T\big(h, z/h\big).$$

Then Ψ has the closed-form derivative

$$\Psi' = \varphi(h)h'\big(2\Phi(z) - 1\big) - \tfrac{1}{\pi}e^{-(h^2+z^2)/2}\,\theta',$$

so (9) splits as $\mathbb{E}[N] = \int_{\mathbb{R}} \Psi' + \int_{\mathbb{R}}(\text{remainder})$, and the boundary values of Ψ give $\int_{\mathbb{R}} \Psi' = 1$ exactly — which is precisely the "1" of (8), i.e. the root every cubic has. What survives is the remainder, and two polynomial identities collapse it: over a common denominator,

$$v + \theta' = \frac{2(x^4 + 6x^2 + 3)}{\sqrt{x^4 + 4x^2 + 1}\,(x^4 + 4x^2 + 9)}, \qquad h^2 + z^2 = \frac{x^4(x^4 + 4x^2 + 9)}{x^4 + 4x^2 + 1}, \tag{10}$$

whereupon $\frac{1}{2}(\mathbb{E}[N] - 1)$ is exactly the integral of Theorem 4. Evenness of the integrand folds $\mathbb{R}$ to $(0, \infty)$. $\square$

Remark 11. θ' is not the nested-radical expression the quotient $(hz' - zh')/(h^2 + z^2)$ suggests: it equals $(x^8 + 6x^6 - 6x^4 - 22x^2 - 3)/\big(q\sqrt{x^4 + 4x^2 + 1}\,(x^4 + 4x^2 + 9)\big)$, the degree-12 denominator of the raw quotient factoring completely. Two consequences: θ' is regular at the origin, with $\theta'(0) = -1/3$, where the quotient form is a genuine $0/0$; and the first identity of (10) is *rational*, reducing to the single polynomial identity $(x^4 + 4x^2 + 1)(x^4 + 4x^2 + 9) + (x^8 + 6x^6 - 6x^4 - 22x^2 - 3) = 2(x^4 + 6x^2 + 3)q$. We initially verified that identity only numerically, believing it to require clearing two nested radicals; the closed form was found while formalizing, and supersedes that step entirely.

5.4. **Verification.** The integrand of Theorem 4 was evaluated to 100 digits and agrees with two structurally unrelated computations of the same probability — a coefficient-space quadrature of the discriminant region and a root-space route — to 79 and 54 digits respectively, each at that method's own precision ceiling. Monte Carlo at 4×10^7 samples sits 0.70σ away. Independently, the Owen's-T cancellation constant $\int_{\mathbb{R}} \Psi'$ evaluates numerically to 1 to 30 digits.

Remark 12. The integral of Theorem 4 is a closed form in the sense that its integrand is elementary and it is a single definite integral — not in the sense of being a finite combination of named constants. A calibrated blind PSLQ search (over 5×10^5 trials against a basis of the constants this problem's siblings produce: π, $\ln 2$, $\ln 3$, $\sqrt{2}$, $\sqrt{3}$ and small rational combinations) found no such combination at 40 digits. That is a negative result about a *named-constant* closed form, and is independent of, and does not contradict, the closed form proved above. We have not verified the natural follow-up claim that the integrand admits no elementary antiderivative: the textbook Liouville criterion does not apply directly, since the integrand is $e^g f$ with g rational but f algebraic, so settling it needs Risch over $\mathbb{Q}(x, \sqrt{x^4 + 4x^2 + 1})$. We record the question as open rather than assert either answer.

6. Formal verification in Lean

All five theorems have been formalized and *completely, machine-checked proved* in Lean 4 against Mathlib (toolchain `v4.33.0`), with **zero** uses of `sorry` anywhere in the development and no unexpected axioms: `#print axioms` on every headline theorem returns exactly `[propext, Classical.choice, Quot.sound]`, the three standard axioms used throughout Mathlib itself. The development is roughly 9000 lines across 27 files, all in the accompanying repository.

The discriminant–root-count bridge. Everywhere above, "has three real roots" is proved as a statement about the *sign of* Δ. Mathlib has `Cubic.discr` and a criterion for `discr` $\neq 0$ implying distinct roots over a splitting field, but nothing connecting the *sign* of a real cubic's discriminant to how many of its roots are real – the fact stated without proof in Section 2 above, and standard since at least Cardano. `DiscriminantRootCount.lean` proves it from scratch and with no appeal to calculus: for $p = b^2 - 3ac > 0$, evaluating $27a^2 f$ at the two points where $3at + b = \mp\sqrt{p}$ gives values whose *product* is $-27a^2\Delta$ and whose *difference* is $4p^{3/2}$, and three applications of the intermediate value theorem to the resulting sign pattern produce three real roots directly, with no derivative or critical point appearing anywhere. Theorems 1 and 2 are additionally restated and proved as `theorem1_root_count` and `theorem2_root_count`, in terms of `HasThreeDistinctRealRoots` directly, with no discriminant in the statement at all.

The Gaussian case without Rice's formula. Theorem 4 is the substantial formalization. Mathlib has no Rice's formula, no coarea formula, and no theory of level crossings of random functions; it does not even have the normal CDF Φ, let alone Owen's T. None of that machinery turned out to be necessary. Because $Z(x)$ is a linear functional of a single fixed three-dimensional Gaussian vector rather than a genuine stochastic process, the entire argument reduces to: the one-dimensional area formula for a cubic (three monotone pieces), Tonelli in $[0, \infty]$-valued integrals, a shear plus a Gaussian integral with a linear term, the folded-normal mean, and Owen's T built from scratch — its definition, both partial derivatives via differentiation under the integral sign, its limits, and the cancellation $\int_{\mathbb{R}} \Psi' = 1$, proved abstractly and then instantiated. The link back to root counts, $f(x_1)f(x_2) = -\Delta_3/27$ and $f(x_1) - f(x_2) = (x_2 - x_1)^3/2$, is two `ring` calls.

The one-sided cube by reuse. Theorem 5, by contrast, was cheap: its four files (roughly 1300 lines) reuse the Theorem 3 cone infrastructure – `IsCone`, the slice-volume law $\mathrm{vol}_3(t \cdot E) = t^3 \,\mathrm{vol}_3(E)$, the coefficient-reversal measure-preservation, and the band lemmas of `Basic.lean`/`Integrals.lean` – almost verbatim, together with `volume_T2Set` $= \frac{1}{2880}$ for the $a = 1$ face. The two genuine differences flagged in §4.4 – four cone pieces rather than eight, and $s \in (0, 1]$ with the band clipping at $s = \frac{1}{2}$ – are exactly the two places where copying the Theorem 3 proof would have produced a false statement, and the formalization pins both down. The headline lemma (named `theorem4` in the source – the Lean development numbers its uniform theorems and does not count the Gaussian case) states `volume` $T4Set$ = `ENNReal.ofReal` $(719/2880 - \log 2/3)$, and `theorem4_root_count` restates it with `HasThreeDistinctRealRoots` and no discriminant.

Simplifications found while formalizing. Three of the informal arguments above turned out to need less machinery than they appear to. In Theorems 1–2 the critical points $x_\pm$ can be skipped entirely in favour of the single `ring`-provable identity $-27\Delta_3(a,b,c) = (27c-9ab+2a^3)^2-4(a^2-3b)^3$, which yields both branches of (3) at once. The cone step (6) needs *no divergence theorem*: slicing $[-1,1]^4$ at $x_1 = t$, using homogeneity to identify the slice with $t\cdot$(face) and integrating $\int_0^1 t^3\,dt = \frac{1}{4}$, is Fubini plus the scaling law $\mathrm{vol}_3(t\cdot E) = t^3\,\mathrm{vol}_3(E)$ — no boundary integral, so the cusps of $\{\Delta = 0\}$ never arise and no smoothness of ∂R is used. And, as noted in §5, θ' has a closed form that makes (10) rational. All three are the same mathematics as the informal derivations, and none changes any number above.

Remark 13. Formalizing surfaced one genuine error, and confirmed that none of the paper's numerical claims were wrong. An early Lean statement of the inner integral $F(s)$ was false exactly at $s = 1$, where $a_0 = 0$ and the integral genuinely diverges; both sides collapsed to unrelated junk values under Lean's total-function conventions ($0/0 = 0$, $\log 0 = 0$) rather than raising an error, and only attempting the proof — not merely writing the statement — caught it. The fix is an $s \neq 1$ hypothesis, harmless since $\{1\}$ is null. An unchecked claim, even one that type-checks, is not yet a fact.

7. Novelty and prior art

An eight-agent parallel literature sweep (arXiv, journal databases, Math StackExchange and MathOverflow's own search indices, OEIS, and general web search), followed by a targeted search on the exact constants derived here, found:

- **Theorem 1**: no prior statement of this probability anywhere searchable (OEIS direct digit-sequence query: no match; Math StackExchange/MathOverflow site search for the monic cubic: zero threads; web search for the exact fraction and decimal: empty). Two caveats. First, the *unnormalized* volume $V_0 = \frac{766}{1215} + \frac{\ln 3}{6}$ (i.e. eight times our value) appears, uninterpreted as a probability, as an intermediate step of the `dxdy.ru` thread discussed above – an unrefereed, partly AI-assisted forum derivation of the *non-monic* cubic, dated June 2026, not indexed by any search engine at the time of our search. Second, Li [1] claims an exact expression for the cubic family $x^3 + 3ax^2 + 3bx + 2c$ with (a,b,c) uniform on symmetric intervals $[-h,h] \times [-k,k] \times [-l,l]$; at $h = k = 1/3$, $l = 1/2$ this specializes to exactly our model. The paper itself is paywalled and, per its recovered *Mathematical Reviews* notice [2], no numerical or symbolic value from it has ever been reproduced by either of its two citations; whether it correctly reduces to Theorem 1 cannot be checked without the original text.
- **Theorem 2**: likely new without caveat – Li's furnished special cases are all symmetric intervals, and the value does not appear in the `dxdy.ru` thread or anywhere else searched.
- **Theorem 3**: the *value* is not new (see above); the *proof* and its independent verification are.
- **Theorem 5**: no prior appearance found. A targeted 2026 search returned zero full-text hits for 719/2880, 479/960, or the decimal on Math StackExchange or MathOverflow (with 1/2880 as a working control returning matches), nothing on OEIS or the general web, and nothing relevant on arXiv; the two known prior threads are both symmetric-interval only. Li's furnished special cases are again all symmetric intervals, so as with Theorem 2 the

one-sided value is not among them – subject to the same standing caveat that Li's text is unread.

- **Theorem 4**: no prior statement found. The Kac–Rice technique is of course standard, and the Gaussian *expected root count* for random polynomials is classical (Kac; Edelman–Kostlan), but we found no source carrying it through to a closed form for the *all-roots-real* probability of the Gaussian cubic, nor the specific integral of Theorem 4 in any searchable source.

We regard the Li [1] question as the one genuinely open item in this account.

8. Open problems

The reduction techniques used here appear to generalize.

(1) **Monic quartic.** The quartic discriminant is homogeneous of degree 6 in the *five* coefficients a, b, c, d, e of $ax^4+bx^3+cx^2+dx+e$, so the divergence-theorem argument of §4 does apply – but to the *non-monic* quartic on $[-1, 1]^5$, reducing that all-real-roots volume to a sum over $2 \times 5 = 10$ three-dimensional face volumes. One of those faces (the one with the leading coefficient pinned to 1) is exactly the *monic* quartic probability we would actually like; the reduction therefore needs it as an *input*, not a consequence, and does not apply directly to the monic problem stated here. Fixing the leading coefficient at 1 breaks the homogeneity the cone argument relies on, so the monic quartic likely needs a direct generalization of the band argument of §3 instead – complicated by a quartic's derivative being itself a cubic, with up to three real critical points rather than one band. Current Monte Carlo estimate: $0.0054749 \pm 9.0 \times 10^{-6}$; under active investigation at the time of writing.

(2) **An antiderivative for Theorem 4.** Does the integrand of Theorem 4 admit an elementary antiderivative, or does the definite integral admit a named-constant closed form? We believe both answers are negative but have verified neither; see the closing remark of §5 for what a proof of the first would require.

(3) **General degree n.** The probability of no real roots for a degree-n Kac polynomial decays as $n^{-3/4}$ exactly (Poplavskyi–Schehr [7]); the corresponding rate constant for *all* n roots real, under speed-n^2 large-deviation scaling, is not known in closed form for uniform (or Kac-Gaussian) coefficients.

(4) **The parent matrix problem.** For an $n \times n$ matrix with i.i.d. uniform entries, the all-real-eigenvalue probability remains open for $n \geq 3$; the striking empirical near-coincidence $\mathbb{P}_n(\text{uniform}) \approx 2^{-(n-1)(n-2)/4}$ (Edelman's Gaussian probability at rank $n-1$), first observed numerically on Math StackExchange, has no known explanation.

Appendix A. Numerical verification summary

	Thm 1	Thm 2	Thm 3	Thm 4	Thm 5
Symbolic (`sympy`, exact)	PASS	PASS	PASS	PASS	PASS
Independent numerical route	$\sim 10^{-16}$	$\sim 10^{-13}$	19 digits	79 digits	2.5×10^{-39}
Blind PSLQ / `identify`	—	—	exact form	(neg.; §5)	exact form
Monte Carlo	$+1.3\sigma$	-0.5σ	-0.73σ	$+0.70\sigma$	$+0.05\sigma$
Lean 4 / Mathlib	**0** `sorry`	**0** `sorry`	**0** `sorry`	**0** `sorry`	**0** `sorry`

Data and code availability

All source code (Python: `sympy`/`mpmath`/`numpy`/`scipy`; Lean 4: Mathlib `v4.33.0`), raw numerical results, the literature-sweep records underlying §7, and an interactive visual explainer of the monic-cubic case are available in the accompanying repository.

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